Sk. Md. Tashfin Sami

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Highly motivated and passionate Computer Science graduate with strong programming knowledge, problem solving experience and practical knowledge of AI integration. Published research related to machine learning. Eager to contribute to AI-driven software development in a collaborative environment.

Education

Bachelor of Science in Computer Science and Engineering *Jan 2020 – Dec 2025*
Patuakhali Science and Technology University, Patuakhali, Bangladesh

- CGPA: 3.25 / 4.00
- Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Software Engineering, Operating System, Computer Networks, Statistics, Machine Learning

Higher Secondary Certificate in Science *Jul 2017 – May 2019*
Khulna Collectorate Public School and College, Khulna, Bangladesh

- GPA: 5.00 / 5.00

Secondary School Certificate in Science *Jan 2015 – Mar 2017*
Khulna Zilla School, Khulna, Bangladesh

- GPA: 4.77 / 5.00

Technical Skills

- **Programming Languages:** C, C++, Java, Python, HTML, CSS, JavaScript
- **Databases:** MySQL
- **Problem Solving:** 250+ problems solved on *Codeforces* , *LeetCode* and *Virtual Judge*
- **AI Integration:** OpenAI API
- **Tools and Platforms:** Git, GitHub, Postman, Linux

Projects

Blood Donation Management System | *Java, Java Swing, MySQL, JDBC* *GitHub*

- Built a desktop application using OOP to structure the system into modular classes for donors, inventory and records management.
- Implemented full CRUD operations for donor management with search and filter functionality by blood group and location.
- Integrated MySQL database via JDBC for persistent storage and designed a user friendly GUI using Java Swing.

Medical Imaging Report Generation Pipeline | *Python, Qwen3, OpenAI API* *GitHub*

- Built a pipeline that converts mammogram images into structured diagnostic reports using a large language model (Qwen3 via OpenAI Chat Completions API).
- Implemented function calling to enforce structured output fields to enforce consistent report format and reduce hallucinations.
- Applied deterministic inference techniques including temperature scaling, nucleus sampling controls, and fixed seed configuration to ensure reproducible and clinically consistent report generation.

Research & Publications

Diffusion Augmented SkinVit-Net for Skin Cancer Diagnosis | *Computer Vision, Generative AI* *IEEE*

- Published at the 28th IEEE International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, December 2025.
- Proposed SkinVit-Net, a lightweight MobileNetV3 Transformer hybrid classifier that achieved **93.52% accuracy** on the HAM10000 skin lesion dataset, outperforming InceptionV3 and Xception baselines.
- Addressed class imbalance in medical imaging by fine tuning a Stable Diffusion 1.5 model with LoRA to generate high quality synthetic dermoscopic images for underrepresented lesion classes.
- Evaluated synthetic image quality using LPIPS, SSIM, and PSNR metrics; model trained on Google Colab T4 GPU demonstrating feasibility in resource constrained environments.

Awards & Scholarships

Ankur Undergraduate Research Scholarship *2025*
Ankur International, Portland, USA

Reference

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